

Isotensor $\pi\pi\pi$ scattering with a ρ resonant subsystem from QCD

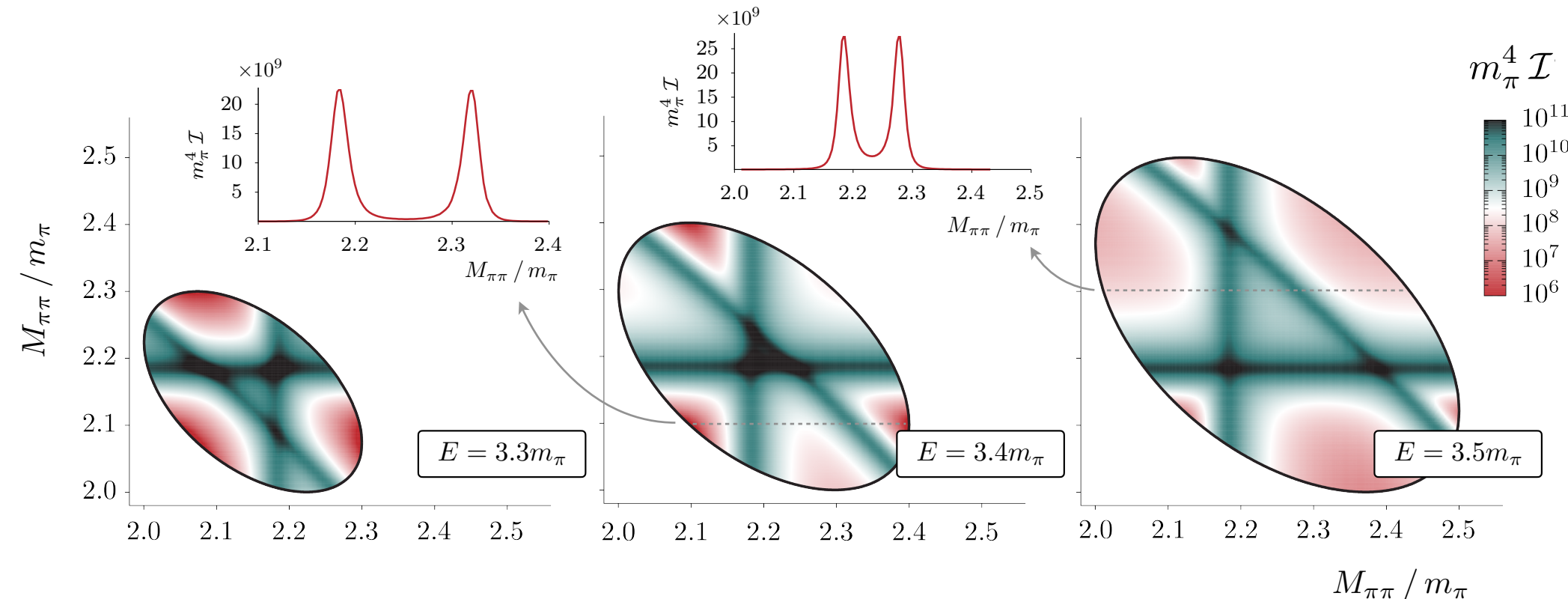
Objectives

- Determine the isotensor $\pi\pi\pi \rightarrow \pi\pi\pi$ scattering amplitude directly from lattice QCD in the $J^P = 1^+$ channel.
- Treat both resonant $\rho\pi$ and nonresonant $\pi\pi\pi$ subchannels in a unified three-body formalism.
- Constrain infinite-volume amplitudes using finite-volume spectra from three lattice volumes together with previously determined $\pi\pi$ amplitudes.

Impact

- First QCD determination of a three-pion amplitude with coupled resonant and nonresonant two-pion subchannels.
- Establishes a practical framework for precision three-hadron spectroscopy and decay studies directly from QCD.
- Opens new paths for applications to unstable hadrons, weak decays, and low-energy nuclear and particle physics.

Dalitz plots of the isotensor three-pion intensity



Representative result: Dalitz-plot intensity distributions for the isotensor $J^P = 1^+$ amplitude at $E/m_\pi = 3.3, 3.4$, and 3.5 . The pronounced structure reflects the resonant $\rho\pi$ subsystem embedded in the fully interacting three-pion amplitude.

Accomplishments

- Computed the isospin-2 three-pion spectrum on three volumes at $m_\pi \approx 400$ MeV, where the ρ is a narrow resonance.
- Combined earlier $I = 1$ and $I = 2$ $\pi\pi$ results with new $\pi\pi\pi$ data to constrain the three-body K matrix.
- Reconstructed the infinite-volume $\pi\pi\pi$ amplitude and resolved genuine three-body effects beyond the $K_3 = 0$ approximation.
- Produced Dalitz-plot intensity predictions for this new class of three-hadron systems.

[Phys.Rev.Lett. 137 \(2026\) 11,111901](#)